Description

Illustration





Figure caption

1. Cross section of particleboard with an outer veneer. 2. Particleboard panels used in house construction. Images used under license from Shutterstock.com

The material

Particle board is fabricated from coarse particles or chips of wood. It has a relatively rough surface best suited as an underlay for veneers and laminates. Low grade material is used as an base for flooring. Higher quality industrial and commercial grade particle boards are stronger and have smoother surfaces, well suited for wall sheathing, general partitioning, shelving and as core stock for laminates such as kitchen tops and cabinets.

Compositional summary

Cellulose/Hemicellulose/Lignin/12%H2O/Adhesive

General properties

Density		450	-	750	kg/m^3
Price	*	0.393	-	0.43	GBP/kg

Material form that data applies to

Bulk	✓
Sheet	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✓
Services	✗

Mechanical properties

Young's modulus	*	1.2	-	2.7	GPa
Shear modulus		0.6	-	1.2	GPa
Bulk modulus	*	1.1	-	2.4	GPa
Bending modulus		2	-	4.5	GPa
Poisson's ratio	*	0.2	-	0.3	
Yield strength (elastic limit)	*	4.2	-	7.2	MPa
Tensile strength		7	-	12	MPa
Compressive strength	*	8	-	16	MPa
Bending strength	*	8	-	15	MPa
Elongation	*	0.29	-	0.35	% strain
Hardness - Vickers	*	3.7	-	4.5	HV
Fatigue strength at 10^7 cycles	*	2.9	-	3.7	MPa
Fracture toughness	*	1	-	2	MPa.m^0.5
Mechanical loss coefficient (tan delta)		0.012	-	0.02	

Thermal and combustion properties

Thermal conductor or insulator?		Good insulator			
Thermal resistivity		7.14	-	8.33	m.°C/W
Thermal expansion coefficient	*	10	-	15	µstrain/°C
Specific heat capacity		1.66e3	-	1.71e3	J/kg.°C
Maximum service temperature		120	-	140	°C
Minimum service temperature		-73	-	-23	°C
Flammability		Highly flammable			
Emissivity		0.81	-	0.9	

Hygro-thermal properties

Water absorption		12	-	35	%
Water vapor permeability		4.99e-12			kg.m/s.m^2.Pa
Frost resistance		Good			

Electrical properties

Electrical conductor or insulator?		Poor insulator			
Electrical resistivity	*	1.05e13	-	2.45e15	µohm.cm
Dielectric constant (relative permittivity)	*	2	-	3	
Dissipation factor (dielectric loss tangent)	*	0.2	-	0.3	
Dielectric strength (dielectric breakdown)	*	1	-	2	MV/m

Optical properties

Natural color

Light brown

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?	No			

Acoustic properties

Sound absorption	Good			
Sound isolation	Average			

Processability (scale 1 = impractical to 5 = excellent)

Moldability	3	-	4	
Machinability	5			

Durability

Water (fresh)	Limited use
Water (salt)	Limited use
Weak acids	Limited use
Strong acids	Unacceptable
Weak alkalis	Acceptable
Strong alkalis	Unacceptable
Organic solvents	Acceptable
UV radiation (sunlight)	Good
Wear resistance	Unacceptable
Industrial atmosphere	Limited use
Rural atmosphere	Limited use
Marine atmosphere	Limited use

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 0.602	-	0.664	kg/kg
Embodied energy (virgin grade production)	* 16.5	-	18.2	MJ/kg
Water Usage	* 665	-	735	l/kg

Material processing: energy

Machining, coarse, embodied energy	* 15.6	-	17.5	MJ/kg
Machining, fine, embodied energy	* 17.7	-	20.2	MJ/kg
Machining, grinding, embodied energy	* 19.8	-	23	MJ/kg

Material processing: climate change

Machining, coarse, climate change (CO2-eq)	* 1.02	-	1.15	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.16	-	1.33	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.29	-	1.51	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.313	-	0.345	kg/kg
Embodied energy, recycling	* 13.7	-	15.1	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 19.7	-	20.7	MJ/kg
Combustion CO2	* 1.73	-	1.82	kg/kg
Landfill	✓			
Biodegrade	✓			
A renewable resource?	✓			

Supporting information

Typical uses

Floor underlays, mobile home decking, shelving, countertops, kitchen cabinets, door core, stair treads, moldings, wall sheathing and siding.

Links

Reference